

Nmap Port States

	State	Meaning	TCP scan trigger (-sS / -sT)	UDP scan trigger (-sU)	Command → packets → output	Pentester takeaway
PRIMARY STATES (the three you'll see in 95% of scans)						
	open	An application is actively listening and accepting connections on that port. This is your attack surface.	SYN-ACK packet received → the 2nd step of the 3-way handshake. Target says "yes, I'm here and willing to talk."	UDP response received from the port. Rare — most UDP services only respond to valid protocol-specific payloads.	TCP SYN SCAN AGAINST OPEN PORT 80: <pre>\$ sudo nmap -sS -p 80 192.168.1.20</pre> Nmap → Target: TCP SYN, dst port 80 Target → Nmap: TCP SYN-ACK, src port 80 ← OPEN Nmap → Target: TCP RST (<i>tears down</i>) Starting Nmap 7.94 (https://nmap.org) at 2026-05-23 14:30 UTC Nmap scan report for 192.168.1.20 Host is up (0.0012s latency). PORT STATE SERVICE 80/tcp open http Nmap done: 1 IP address (1 host up) scanned in 0.34s UDP SCAN AGAINST OPEN PORT 53 (DNS): <pre>\$ sudo nmap -sU -p 53 192.168.1.20</pre> Nmap → Target: UDP DNS query, dst port 53 Target → Nmap: UDP DNS response ← OPEN PORT STATE SERVICE 53/udp open domain	Every open port is an opportunity. Always follow up with <code>-sV</code> (version detection) and <code>-sC</code> (default scripts). Knowing port 22 is open is meaningless — knowing it's <i>OpenSSH 7.2p2</i> is actionable.
	closed	The host is alive and reachable, but nothing is listening on that specific port. The OS actively rejected your probe per RFC 793.	RST (reset) packet received. The target's TCP/IP stack got the SYN, found no listener, and replied "nobody home."	ICMP "Port Unreachable" (type 3, code 3). Kernel received the UDP packet, found no listener, sent back ICMP error.	TCP SYN SCAN AGAINST CLOSED PORT 81: <pre>\$ sudo nmap -sS -p 81 192.168.1.20</pre> Nmap → Target: TCP SYN, dst port 81 Target → Nmap: TCP RST ← CLOSED Nmap scan report for 192.168.1.20 Host is up (0.0011s latency). PORT STATE SERVICE 81/tcp closed hosts2-ns Nmap done: 1 IP address (1 host up) scanned in 0.28s UDP SCAN AGAINST CLOSED PORT 161: <pre>\$ sudo nmap -sU -p 161 192.168.1.20</pre> Nmap → Target: UDP, dst port 161 Target → Nmap: ICMP type 3, code 3 ← CLOSED (port unreachable) PORT STATE SERVICE 161/udp closed snmp	Still useful info: confirms the host exists, gives baseline for OS fingerprinting, and tells you no firewall is in the way . Lots of closed ports + a few open = an unfirewalled host with a small attack surface.
	filtered	A firewall, ACL, or packet filter is interfering. Nmap can't determine if the port is open or closed — something dropped or blocked the probe.	No response after retransmissions, OR ICMP unreachable with codes 1, 2, 9, 10, or 13 (admin prohibited, etc.).	Same as TCP: no response after retries, or ICMP unreachable with codes other than 3.	SYN SCAN AGAINST FIREWALLED PORT 22 (SILENT DROP): <pre>\$ sudo nmap -sS -p 22 192.168.1.20</pre> Nmap → Target: TCP SYN, dst port 22 (nothing – silently dropped) Nmap → Target: TCP SYN (retransmit) (still nothing) ← FILTERED Nmap scan report for 192.168.1.20 Host is up (0.0014s latency). PORT STATE SERVICE 22/tcp filtered ssh Nmap done: 1 IP address (1 host up) scanned in 2.14s FIREWALL EXPLICITLY BLOCKING (WITH --REASON): <pre>\$ sudo nmap -sS --reason -p 22 192.168.1.20</pre>	Most ambiguous result. If every port shows filtered, you're behind a default-deny firewall. Try: different source IP, <code>--source-port 53</code> , fragmentation (<code>-f</code>), slower timing (<code>-T1</code>), or pivot internally. Filtered = no answer about the port, but information about the firewall policy .

	State	Meaning	TCP scan trigger (-sS / -sT)	UDP scan trigger (-sU)	Command → packets → output	Pentester takeaway
					Nmap → Target: TCP SYN, dst port 22 Firewall → Nmap: ICMP type 3, code 13 ← FILTERED (admin prohibited) PORT STATE SERVICE REASON 22/tcp filtered ssh admin-prohibited from 192.168.1.1	
SECONDARY STATES (situational — know they exist)						
	open filtered	Nmap can't tell whether the port is open or filtered. Almost always a UDP scan artifact, also seen in -sN / -sF / -sX stealth scans.	Only appears with Null (-sN), FIN (-sF), and Xmas (-sX) scans when there's no response.	No response at all. UDP is connectionless — an open port with nothing to say looks identical to a dropped packet.	UDP SCAN AGAINST PORT 161 (SNMP), NO COMMUNITY STRING: \$ sudo nmap -sU -p 161 192.168.1.20 Nmap → Target: UDP probe, dst port 161 (no response – could be open and silent OR firewall dropped the packet) ← OPEN FILTERED Nmap scan report for 192.168.1.20 Host is up (0.0013s latency). PORT STATE SERVICE 161/udp open filtered snmp Nmap done: 1 IP address (1 host up) scanned in 4.31s RESOLVE WITH VERSION DETECTION: \$ sudo nmap -sU -sV -p 161 192.168.1.20 Sends protocol-specific SNMP probes that coax a real response → resolves ambiguity PORT STATE SERVICE VERSION 161/udp open snmp SNMPv1; net-snmp v3	Don't report these as findings without verification. Always re-scan with -sV to send protocol-specific probes (DNS query, SNMP GetRequest, NTP request) that force real services to respond.
	closed filtered	Nmap can't tell whether the port is closed or filtered. Only appears in IP ID idle scans (-sI) — an artifact of inferring state indirectly through a zombie host.	Only with -sI <zombie> idle scan. The IP ID inference can't always distinguish closed from filtered.	N/A — idle scans are TCP-only.	IDLE SCAN USING A ZOMBIE HOST: \$ sudo nmap -sI 192.168.1.50 192.168.1.20 Nmap probes zombie's IP ID before → X Nmap spoofs SYN from zombie to target Nmap probes zombie's IP ID after → X+1 (IP ID didn't jump by 2 – ambiguous) ← CLOSED FILTERED Idle scan using zombie 192.168.1.50 (192.168.1.50:80); Class: Incremental Nmap scan report for 192.168.1.20 Host is up (0.052s latency). Not shown: 998 closed filtered ports PORT STATE SERVICE 80/tcp open http 443/tcp open https Nmap done: 1 IP address (1 host up) scanned in 28.46s	Rare in practice. If you're using idle scans you already know what you're doing. Mostly relevant for true stealth engagements where attribution must be avoided. For most pentests, you'll never see this state.
	unfiltered	The port is reachable (no firewall blocking), but Nmap can't determine if a service is listening. Only appears in ACK scans (-sA), which map firewall rules — not port states.	RST received in response to an ACK probe → packet got through, so the port isn't filtered. But ACK probes can't reveal open/closed.	N/A — ACK scans are TCP-only.	ACK SCAN TO MAP FIREWALL BEHAVIOR: \$ sudo nmap -sA -p 22,80,443 192.168.1.20 Nmap → Target: TCP ACK, port 80 Target → Nmap: TCP RST + UNFILTERED (firewall let it through) Nmap → Target: TCP ACK, port 443 (no response – firewall blocked) ← FILTERED Nmap scan report for 192.168.1.20 Host is up (0.0014s latency). PORT STATE SERVICE 22/tcp filtered ssh 80/tcp unfiltered http 443/tcp filtered https Nmap done: 1 IP address (1 host up) scanned in 1.42s	ACK scans are for firewall reconnaissance , not service discovery. Combine with -sS : SYN scan reveals open/closed, ACK scan reveals filtered/unfiltered. Together = complete picture of the network's filtering policy.